

**DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE**

**Supplementary - Winter 23**

**Course: B. Tech. Branch :** Computer Engineering/Computer Science & Engineering

**Subject Code & Name:** BTCOE604 (B)\_Y23 Internet of Things

**Semester :** VI

**Max Marks:** 60

**Date:** 23-01-24

**Duration:** 3 Hr.

**Instructions to the Students:**

1. All the questions are compulsory.
2. Draw neat diagram wherever necessary.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

(Level/CO) Marks

**Q.1 Solve Any Two of the following.**

- |                                                                     |             |    |
|---------------------------------------------------------------------|-------------|----|
| A) What is IoT? Explain its characteristics and application of IoT. | Application | 06 |
| B) What are the biggest challenges for IoT adoption?                | Understand  | 06 |
| C) Explain with neat diagram different IoT communication models.    | Remember    | 06 |

**Q.2 Solve Any Two of the following.**

- |                                                         |            |    |
|---------------------------------------------------------|------------|----|
| A) Explain communication criteria for an IoT platforms. | Understand | 06 |
| B) What is actuator? Give classification of actuators.  | Analysis   | 06 |
| C) Explain leading types of IoT wireless technologies.  | Understand | 06 |

**Q.3 Solve Any Two of the following.**

- |                                                               |            |    |
|---------------------------------------------------------------|------------|----|
| A) List and explain key advantages of Internet protocol (IP). | Understand | 06 |
| B) Describe Message Queuing Telemetry Transport Protocol.     | Understand | 06 |
| C) Explain supervisory control and data acquisition.          | Understand | 06 |

**Q.4 Solve Any Two of the following.**

- |                                                                            |            |    |
|----------------------------------------------------------------------------|------------|----|
| A) What is machine learning? Explain types of machine learning categories. | Remember   | 06 |
| B) What is Edge analytics? Describe Edge Analytics Core Functions.         | Understand | 06 |
| C) Explain Massively Parallel Processing Shared-Nothing Architecture.      | Understand | 06 |

**Q.5 Solve Any Two of the following.**

- |                                                                  |             |    |
|------------------------------------------------------------------|-------------|----|
| A) Explain smart city IoT architecture with a neat diagram.      | Understand  | 06 |
| B) Explain smart city security architecture with a neat diagram. | Understand  | 06 |
| C) Discuss an IoT Strategy for Smarter Cities.                   | Application | 06 |

\*\*\* End \*\*\*